Bump Once, Remember Wisely: Calibrated IMU Collision Detection and Evidence-Aware, Forgetting Collision Memory for Mobile Robots Among Invisible and Moving Obstacles

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I. Problem Description and Primary Reference Paper

Problem. LiDAR-based navigation assumes that every obstacle is seen by the LiDAR. Two common cases break this assumption: (1) *invisible obstacles*—glass walls and doors, transparent curtains and thin objects let laser beams pass through, so the map shows free space and the planner drives into them; and (2) *obstacles that enter unseen*, e.g., from the blind spot behind a 270° LiDAR. In both cases a collision cannot be avoided from perception alone. The robot must therefore (i) *detect* that it collided, (ii) *react* safely, and (iii) *remember* the obstacle so that it does not hit it again. Remembering creates a new difficulty: if the obstacle was a person, a cart that is later moved, or a door that later opens, a permanent memory turns it into a *phantom wall*. A *false* detection does the same and can make the goal unreachable. **Our problem** is to detect collisions with a *known* false-alarm rate, and to remember them selectively: keep invisible static obstacles, forget transient ones and false alarms.

Primary reference paper. R. Wang *et al.*, "Air Bumper: A Collision Detection and Reaction Framework for Autonomous MAV Navigation," *IEEE ICRA 2024*, pp. 15735–15741 [1]. It is an original research paper from an approved venue (ICRA) in the required period. A caged drone with a 3D LiDAR uses only its IMU to: *detect* a collision when any body-axis acceleration exceeds a fixed threshold ($a^* = 20\,\mathrm{m/s^2}$, sliding window $N = 10$); *estimate* its direction and contact point assuming a spherical cage; *react* immediately by moving away from the hit, blended with the gradient of the distance to known obstacles; and *remember* it by inserting the contact patch into the volumetric map as a "collision sensor" whose marks are *never cleared* by the LiDAR. Recovery from glass doors is shown in Gazebo, and from glass and stick hits on a real drone (detection $\approx$0.18 s, recovery $<$1 s, 10 trials).

Why this paper. It is one of the first frameworks to combine collision detection, reaction *and* mapping in a standard LiDAR navigation stack using only the IMU. It turns failures into map knowledge ("failure as a sensor"), which is directly a *failure detection and recovery* problem. It is open source [2] and validated in simulation and on hardware. Its design choices are simple and explicit (fixed threshold, permanent marks, spherical body), so their limitations can be analyzed and improved within a term project. Related work handles failures differently: [3] predicts planner failures *before* they happen from LiDAR geometry, and [4] adapts edge costs online from contact feedback; both assume the LiDAR sees every obstacle, which is exactly what fails with glass.

TABLE I
Software and simulation environment (all free).

Component	Choice	Justification
Middleware	ROS 2 Jazzy [6] in Docker	LTS, free; identical setup on every laptop
Navigation	Nav2 [7]: Smac/NavFn planner, MPPI controller, layered costmaps [8] with keepout and speed *costmap filters*	Industry-standard stack; filters inject memory *without modifying Nav2*
Simulator	Gazebo Harmonic [9], headless batch runs	Same family as [1]; free ground truth
Robot	Clearpath Jackal (diff drive, 0.51×0.43 m), 270° 2D LiDAR, IMU (100–200 Hz, noise + bias), wheel odometry, chassis contact sensor	Common research AMR; contact sensor gives ground-truth labels and is *not* used by the detector
Invisible obstacles	Models with collision but no visual geometry	GPU LiDAR renders visuals only, so beams pass through while physics blocks: glass as in [1]
Moving / transient	Joint/trajectory-driven boxes and carts, doors opening at random times, objects removed after contact	Creates "remembered but gone" cases

Relevance beyond drones. Glass is common in offices, airports, malls and hospitals. Ground AMRs there mostly rely on bumper switches that only *stop* the robot and remember nothing [5]. We therefore study the Air Bumper pipeline on a *ground robot*, where detect–react–remember appears with new failure modes: the robot's own braking, floor bumps, and a box-shaped body.

II. Environment, Benchmark, and Dataset

Table I summarizes the environment. Because no public benchmark contains invisible and transient obstacles for ground

robots, we build a procedurally generated, seeded **scenario suite**:

- **S1 – Static glass** (60 worlds): office/warehouse layouts with 1–4 glass walls, glass doors (as in [1]) and transparent partitions at random positions. Tests detection and memory on invisible static obstacles.
- **S2 – Transient** (45): (a) a glass door that opens 20–120 s after the first contact; (b) a visible cart blocking the shortest route, removed after contact; (c) a box crossing from the LiDAR blind spot (the analog of the paper's stick test). Tests the phantom-wall problem.
- **S3 – False-alarm stress** (30): *no collisions*; 2–4 cm floor bumps and cable ramps, rough patches, aggressive stop-and-go goals, doubled IMU noise. Measures false alarms per hour.
- **S4 – BARN-Glass** (50, stretch): BARN worlds [10] with 10–30% of obstacles made invisible.

Worlds are split 50/20/30% into train/calibration/test per set; S3 terrain types unseen during calibration form an out-of-distribution test. Speeds $v_{\max} \in \{0.5, 1.0, 1.5\}$ m/s. We plan $\sim$10 h of collision-free driving (for calibration, Sec. IV) and $\sim$2,000 labeled collisions, which fits into a few overnight runs with 2–3 parallel containers.

III. Challenges and Limitations

Table II lists the limitations of [1] that motivate our project, each with evidence from the paper.

These issues matter because a failure detector is only deployable when its false-alarm and miss rates are known, and because a map that never forgets turns every detection error and every moved object into a lasting navigation failure. The open questions are: *Can collision detection have a guaranteed false-alarm rate instead of a hand-tuned threshold? Can collision memory be selective, keeping invisible static obstacles while forgetting transient ones and its own mistakes?*

TABLE II
Limitations of Air Bumper [1].

#	Limitation (evidence)	Why it matters
L1	Fixed, hand-tuned threshold on raw acceleration, $a^* = 20\,\text{m/s}^2$ (Sec. IV-A, "experimentally determined"); no false-alarm or miss rates reported	The robot's *own* braking, turns and floor bumps also cause spikes, while soft or low-speed hits stay below a^*. The false-alarm rate is unknown and changes with platform, speed and floor
L2	Per-axis test $\lvert a_x\rvert > a^*$ *or* $\lvert a_y\rvert > a^*$ (Sec. IV-A)	A diagonal hit splits between axes: 25 m/s^2 at 45° gives 17.7 m/s^2 per axis and is *missed*
L3	Direction from raw acceleration, $\mathbf{a}_c = -\mathbf{a}$ (Eqs. 1–4)	Commanded acceleration biases the direction, especially while braking or turning
L4	Spherical contact model $\mathbf{c}_c = \hat{\mathbf{C}}\, l$ (Eq. 5)	Wrong contact point and wrong mapped cells for a box-shaped robot
L5	Permanent, top-priority marks: "the robot remembers all obstacles it ever collides with" (Sec. V-C)	Right for static glass; wrong for people, moved carts, opened doors and *false detections*, which become phantom walls. Detection errors (L1) are made *permanent* by the map
L6	LiDAR evidence at the contact is ignored	Whether the LiDAR *saw* the obstacle separates visible (already mapped, maybe moving) from invisible (must be remembered)
L7	Small, qualitative evaluation: 1 simulated world, 10 real trials	No TPR, false alarms/hour, latency or direction error; no baselines or statistics

IV. Proposed Method and Novel Contributions

Baseline reimplementation (B1, not a contribution). We first port Air Bumper to ROS 2 for a ground robot: the same per-axis detector, sphere contact model and permanent marks (written as lethal cells into a Nav2 keepout mask). Since a diff-drive robot cannot move sideways, the reaction backs off along the projection of $-\hat{\mathbf{C}}$ on the heading, or rotates away and then moves, for $R_d = 0.3$ m, through a high-priority `twist_mux` channel so that it does not wait for the planner.

A. C1: Model-aware, calibrated collision detection (L1–L4)

(a) Residual signal. Instead of raw acceleration we use the *unexpected* acceleration

$$\mathbf{r}(t) = \mathbf{a}_{xy}^{\text{IMU}}(t) - \hat{\mathbf{a}}_{xy}(t), \tag{1}$$

where $\hat{\mathbf{a}}$ is the acceleration explained by the robot's own commands: $(v_{\text{cmd}}, \omega_{\text{cmd}})$ is passed through a first-order response model identified on collision-free data, giving $\hat{a}_x = \dot{\hat{v}}$ and $\hat{a}_y = \hat{v}\hat{\omega}$. Own braking and turning are thus removed (L1, L3).

(b) Features and score. Over a sliding window $W = 0.2$ s we compute the peak horizontal norm $\|\mathbf{r}_{xy}\|$ (removes the per-axis issue, L2), the peak jerk, the vertical residual and pitch rate (separate floor bumps from collisions), and the tracking error between v_{cmd} and odometry speed (pushing against an obstacle). A small, interpretable logistic regression maps them to a score $s \in [0, 1]$; as an ablation, $s = \|\mathbf{r}_{xy}\|$.

(c) Calibrated threshold. We set the threshold by split conformal calibration [11], [12] on n *collision-free* windows from the calibration worlds:

$$\hat{\tau} = s_{(\lceil (n+1)(1-\alpha) \rceil)}, \tag{2}$$

the $\lceil (n+1)(1-\alpha) \rceil$-th smallest score. Under exchangeability a new collision-free window triggers with probability $\leq \alpha$, so the false-alarm rate is fixed in advance: $\approx \alpha \cdot 3600/W$ per hour for non-overlapping windows. For a target of 1 false alarm/h with $W = 0.2$ s, $\alpha \approx 5.6 \cdot 10^{-5}$ and $n \geq 1/\alpha \approx 18{,}000$ windows ($\approx$1 h of driving; we collect $\sim$10 h). We target $\{0.5, 1, 5\}$ false alarms/h and, since windows are temporally correlated, also report the empirical rate.

(d) Footprint-aware estimation. The direction $\phi = \text{atan2}(-r_y, -r_x)$ is computed from the residual, and the contact point is the intersection of this ray with the robot's *footprint polygon* instead of a sphere (L4). The mapped patch is a 0.3 m segment perpendicular to ϕ.

B. C2: Evidence-aware, forgetting collision memory (L5, L6)

(a) LiDAR evidence. At detection we inspect the rays within $\pm 10^\circ$ of the contact bearing and label the contact *visible* (a return within contact distance $+0.15$ m: already mapped, possibly a person or cart), *invisible* (beams pass beyond the contact: the glass case), or *unseen* (bearing in the blind spot).

(b) Decaying belief. Each mark k stores its position, extent, hit count h_k and last-hit time t_k, with belief

$$b_k(t) = \exp\big(-(t - t_k)/\tau_{\text{class}}\big), \tag{3}$$

$\tau_{\text{invisible}} = 600\,\text{s}$, $\tau_{\text{unseen}} = 60\,\text{s}$, $\tau_{\text{visible}} = 10\,\text{s}$ (enough to prevent an immediate re-hit; the LiDAR covers the rest). All values are tuned on calibration worlds and studied for sensitivity. A new hit within 0.3 m merges with the mark ($h_k{+}1$, t_k reset): a real glass wall is hit again and stays remembered, while a transient obstacle or a false alarm fades.

(c) Graded costs, not walls. Marks are published as two Nav2 costmap-filter masks: a *keepout* mask with cost $c = c_{\max} b_k$, lethal *only if* $h_k \geq 2$ (confirmed) and otherwise high but passable, so a single unconfirmed mark can never make the goal unreachable; and a *speed* mask limiting speed to 0.3 m/s near decayed or unconfirmed marks, so that re-testing a remembered spot causes at most a gentle contact that C1 detects quickly.

Implementation. All nodes are Python (`rclpy`): `imu_residual`, `collision_detector`, `reaction`, `collision_memory`. Memory enters only through standard Nav2 filters, so the method is planner-agnostic. **Expected improvements over [1]:** (1) a false-alarm rate fixed before deployment and verified on test worlds; (2) detection of diagonal and low-speed contacts; (3) smaller direction and contact-point errors on a non-spherical body; (4) as few repeated collisions on static glass as permanent memory, but no phantom walls with transient obstacles or false alarms.

V. Research Questions and Evaluation Plan

RQ1 (Detection): Does C1 achieve a higher true-positive rate than Air Bumper's fixed per-axis threshold *at the same false-alarm rate*, across speeds and floors, and does the calibrated rate hold on unseen terrain? **RQ2 (Estimation):** How much does residual-based, footprint-aware estimation reduce direction and contact-point errors? **RQ3 (Memory):** Does C2 keep repeated collisions with static invisible obstacles as low as permanent memory, while avoiding phantom-wall failures with transient obstacles and false alarms? **RQ4 (System):** Does the full system improve success, collisions and time-to-goal over the baselines?

Baselines are listed in Table III. **Metrics.** *Detection* (per collision event, ground truth from the contact sensor): TPR, false alarms per hour of collision-free driving, empirical vs. target false-alarm rate, detection latency, ROC-AUC. *Estimation:* direction error (deg), contact-point error (m). *Memory:* phantom area (m^2 of marked cost with no real obstacle, integrated over time), repeated collisions (contact within 0.5 m of an earlier one), blocked-goal events. *System:* success rate, collisions per run, time-to-goal, SPL [13], node run time (<5 ms per cycle).

TABLE III
Baselines and ablations.

ID	Detection	Memory	Represents
B0	none (Nav2 LiDAR collision monitor, default recoveries)	none	Standard Nav2
B1	Air Bumper [1]: $a^* = 20\,\text{m/s}^2$ and a^* re-tuned on calibration worlds	permanent, lethal	Primary paper
B2	Oracle bumper (contact sensor)	permanent	Industry bumper; isolates memory
B3	C1	permanent	C1 alone
B4	Air Bumper	C2	C2 alone
Ours	C1	C2	Full method

TABLE IV
Planned experiments.

#	Experiment	Data	RQ
E1	Offline detection: ROC, TPR at matched false-alarm rate; features vs. $\|\mathbf{r}\|$ vs. raw acceleration; per speed	S1–S3 test logs	RQ1
E2	Calibration validity: target $\{0.5, 1, 5\}$/h vs. measured; in-distribution vs. unseen terrain	S3, S1 free segments	RQ1
E3	Estimation: sphere vs. footprint; raw vs. residual	S1, S2	RQ2
E4	Memory (closed loop, oracle detector): permanent vs. decay-only vs. C2	S1, S2	RQ3
E5	Full comparison B0–B4 vs. Ours $\times$ 3 speeds $\times$ 3 seeds	S1–S3 (+S4)	RQ4
E6	Ablations/sensitivity: τ, lethal-after-h, speed mask, LiDAR evidence, W	Calibration worlds	RQ3, RQ4

Experiments are listed in Table IV. **Statistics.** The world is the unit of analysis: 95% bootstrap confidence intervals over worlds, McNemar tests for success/failure and Wilcoxon signed-rank tests for continuous metrics, with Holm correction. All hyperparameters, including the B1 threshold, are tuned only on calibration worlds. **Threats to validity.** Simulated contacts and IMU are idealized, so we add noise/bias and report results per noise level; the conformal guarantee is approximate under temporal correlation, so we report the empirical rate; sim-to-real transfer is not tested.

VI. Team Roles and Individual Contributions

Roles are given in Table V. Each member writes and presents the part for their own module; contributions are documented by per-member git history and a weekly `CONTRIBUTIONS.md`. **Timeline:** week 1 (11–16 Nov) stack running and verification of the invisible-obstacle model and IMU collision spikes; week 2 S1–S3 and B1; week 3 data collection and command-response model; week 4 C1, E1–E3; week 5 C2, E4; week 6 E5, E6, statistics; week 7 report and slides; presentation on 30.12.2026. **Scope:** minimum = B1 + C1 + E1–E3; target = + C2, E4–E5; stretch = S4 and a low-speed contact study.

References

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TABLE V
TEAM ROLES.

Member	Role	Responsibilities and deliverables
[Member 1]	Simulation & scenarios	Docker/ROS 2/Nav2/Gazebo bring-up; Jackal with IMU, LiDAR, contact sensor; invisible-obstacle models; S1–S4 generators; episode runner, logging, ground-truth labels
[Member 2]	Detection & estimation (C1)	Command-response model, residual node, features, logistic regression, conformal calibration, footprint contact estimation; E1–E3
[Member 3]	Reaction & memory (C2)	Diff-drive reaction node, collision-memory node, keepout/speed masks, LiDAR evidence check; E4, E6
[Member 4]	Baselines & evaluation	Air Bumper reimplementation (B1), B0/B2–B4, metrics, batch experiments E5, statistics, figures

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